

```
In [1]: # Behzad Ghabaei
        # CS 82A - Data Science
        # Module 2 Lab 1
        # Python Warm - Up
        # Instructor Seno
        # 9/8/2026
```

Step 1: Wages with f-strings

```
In [2]: name = "Maria-Anna Chen"
        hourly_wage = 16.00
        hours = 40
        weekly_pay = hourly_wage * hours
        print(f"Maria-Anna earns {weekly_pay} per week") #640.0
```

Maria-Anna earns 640.0 per week

Step 2: Count the warm days

```
In [3]: temps = [61,65,72,58,70,74,68]
        high = 0
        for s in temps:
            if s > 65:
                high = high + 1
        print(high) #4
```

4

Step 3: Write a temperature converter

```
In [4]: def to_celcius(f):
        return (f - 32) * 5/9

        print(round(to_celcius(temps[0]), 1) ) #16.1
```

16.1

Step 4: A movie dictionary

```
In [5]: # {"key": "value"}
        book = {"title": "Dune", "year": 1965, "rating": 100}
        for key, value in book.items():
            print(key, ":", value)
```

title : Dune
year : 1965
rating : 100

Step 5: Load and inspect chickwts.csv

```
In [6]: import pandas as pd
        df = pd.read_csv("chickwts.csv") # df = pd.read_csv("chickwts.csv", index_col = 0 )
        df.head()
        # df.tail()
```

```
Out[6]:
```

	Unnamed: 0	weight	feed
0	1	179	horsebean
1	2	160	horsebean
2	3	136	horsebean
3	4	227	horsebean
4	5	217	horsebean

```
In [ ]:
```

`df.shape` would return (71, 3) with the code above written simply as: `df = pd.read_csv("chickwts.csv")`. This includes the Unnamed column. When using the search box from "10 minutes to Pandas" and asking: "How to remove the Unnamed column?" The search returns, under "10 tools" the section called, "Index columns and trailing delimiters." Here is the snippet provided:

```
In [94]: data = "index,a,b,c\n4,apple,bat,5.7\n8,orange,cow,10"
```

```
In [95]: pd.read_csv(StringIO(data), index_col=0)
```

```
Out[95]:
```

	a	b	c
index			
4	apple	bat	5.7
8	orange	cow	10.0

When `index_col = 0` is used in the parameters for data frame (df), such as: `"df = pd.read_csv("chickwts.csv", index_col = 0 )"` The command `"df.shape"` thus returns (71, 2).

```
In [7]: df.shape
```

```
Out[7]: (71, 3)
```

```
In [8]: df.describe()
```

```
Out[8]:
```

	Unnamed: 0	weight
count	71.000000	71.000000
mean	36.000000	261.309859
std	20.639767	78.073700
min	1.000000	108.000000
25%	18.500000	204.500000
50%	36.000000	258.000000
75%	53.500000	323.500000
max	71.000000	423.000000

Step 6: Interpret `describe()`

The **describe()** output tells us a lot about the chick weight csv file. This is from the **Pandas** library. The **count** tells us how many chickens are participating in the survey. The **mean** tells us the average weight in grams of all the 71 chickens combined. The **std** is the standard deviation of all the chickens which describes the spread in between weights. A high std says the data is spread out. The **min** is the lowest value in the data set in chicken weight (grams). The **max** is the highest weight in the dataset. **10 minutes To Pandas** explains the **describe()** function, under "Viewing Data." Describe shows a quick statistic summary of your data. This reference is called "pandas.DataFrame.describe" and gives the central tendency, dispersion, and shape of dataset distribution, excluding NaN values. For numeric data, the results will include: count, mean, std, min, max, and lower percentile as 25, upper percentile as 75, and 50 percentile is the same as the median. ([pandas.pydata.org | https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.describe.html#pandas.DataFrame.describe](https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.describe.html#pandas.DataFrame.describe))

StackOverflow.com explains, imagine sorting your data from lowest to highest. **25%** means 25% of your data have the value 204.5 or below (less than or equal to). **50%** means, 50% of your data have the value 258.0 or below, and **75%** means, **75%** of your data have a value of 323.5 or below. (My Prompt: "What does 25%, 50%, 75% mean in the describe() function?" | "What are 25%,50%,75% values when we describe a grouped dataframe?" | Sibbir Ahmed | <https://stackoverflow.com/questions/57869926/what-are-25-50-75-values-when-we-describe-a-grouped-dataframe>)

End Of Lab - Behzad Ghabaei

In []: